Homework 4

Discrete Math Fall 2023

Due: Thursday, September 14$^{\text{th}}$ at the beginning of class.

Read Section 0.3 of Levin before you start this homework.

(1) Let $A = \{1, 2, 3, 4, 10\}, B = \{3, 4, 5, 6, 7\}$, and $C = \{2, 4, 6, 8, 10\}$ in the universe $\mathcal{U} = \{n \in \mathbb{N} \mid 1 \leq n \leq 10\}$. For each of the following, provide the set in question as a list.

(a) Find $A \cup B$.
(b) Find $A \cap B$.
(c) Find $A \cup C$.
(d) Find $A \cap C$.
(e) Find $B \cup C$.
(f) Find $B \cap C$.
(g) Find $A \setminus B$.
(h) Find $A \setminus C$.
(i) Find $B \setminus A$.
(j) Find $B \setminus C$.
(k) Find $C \setminus A$.
(l) Find $C \setminus B$.
(m) Find $\overline{A}$.
(n) Find $\overline{B}$.
(o) Find $\overline{C}$.
(p) Find $\overline{A \cup C}$.
(q) Find $A \setminus (B \cap C)$.
(r) Find $A \cup (B \cap C)$.
(s) Find $(A \cup B) \cap (A \cup C)$.
(t) Find $A \cap (B \cup C)$.

(2) Let $A = \{1, 3, 4, 8, 10\}, B = \{n \in \mathbb{N} \mid 4 \leq n \leq 15\}$, and
$C = \{n \in \mathbb{N} \mid n = 2m - 1, m \in \mathbb{N}\}$.

(a) Find $B \setminus A$.
(b) Find $B \setminus C$.
(c) Find $A \setminus B$.
(d) Find $A \setminus C$.
(e) Find $A \cap B$.
(f) Find $A \cap C$.
(g) Find $B \cap C$.
(h) Find $A \cup B$.

(3) Draw a Venn Diagram to represent each of the following sets. Be sure to draw a box around them to represent the universe.

(a) $B \setminus A$.
(b) $B \cap \overline{A}$.
(c) $B \cap A$.
(d) $A \cup B$.
(e) $\overline{A} \cap B$.
(f) $A \cap B \cap C$.
(g) $(A \cup B) \cap C$.
(h) $A \cup (B \cap C)$.

(4) Let $A = \{1, 2, 3, 4\}, B = \{2, 4, 6\}$, and $C = \{a, b, c\}$.

(a) Write out the following sets.

(i) $A \times B$.	(iii) $C \times B$.
(ii) $B \times A$.	(iv) $C \times C$.

(b) What are the cardinalities of the sets in part (a)?

(c) Does $A \times B = B \times A$. Explain.

(d) Find $(A \times B) \cap (B \times A)$.

(e) What is the general rule for $|X \times Y|$?

(5) For any sets A and B, define $A + B = \{a + b : a \in A \wedge b \in B\}$. For the following, what is $A \cup B$ and what is $A + B$?

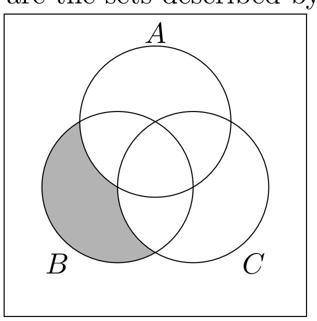
(a) $A = \{1, 2\}$ and $B = \{2, 3, 4\}$

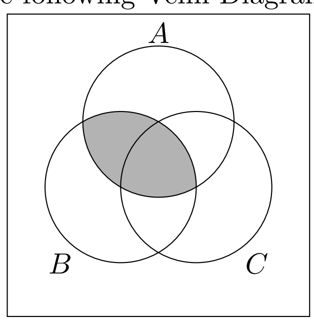
(b) $A = \{0, 5, 7\}$ and $B = \{22, 41\}$

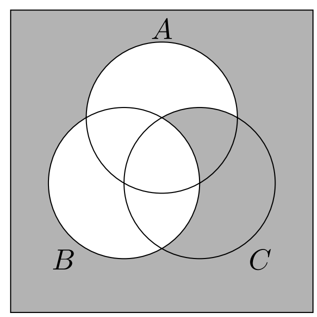
(c) $A = B = \{0, 1\}$

(d) If A= $\{a, b\}$ and $B = \{a, b, c\}$ for $a, b, c \in \mathbb{R}$

(6) What are the sets described by the following Venn Diagrams?

(a) 

(b) 

(c) 

(7) Let $A = \{2, 4, 6, 8\}$. Suppose B is a set with $|B| = 5$.

(a) What are the smallest and largest possible values of $|A \cup B|$? Explain.

(b) What are the smallest and largest possible values of $|A \cap B|$? Explain.

(c) What are the smallest and largest possible values of $|A \times B|$? Explain.